Revise on August 26 and 27

UnionLabs — X310 transmit failure: diagnosis and changes

Merged to `main` as `290de1e`

Summary. An X310 link synchronised perfectly on every burst and then failed CRC on every frame, in every modulation scheme, at every gain setting. The cause was a single symbol lost at a fixed offset inside each burst, introduced by the transmit send loop. Nine commits: one is the fix, four made it findable, and the rest are genuine defects found on the way. Verified on hardware — both `linktest` and `radio.sh` now decode every chunk.

The evidence that located it

The receiver was changed to report *where* the coded bits stop matching a known transmission, rather than averaging the disagreement into a single number. Forcing the transmit chunk size then moved the fault by exactly the amount the boundary moved:

`--tx-spb`	first bad coded bit	data offset	chunk boundary	implied lead
1996 (device default)	911	1822 samples	1996	**174**
1000 (forced)	414	828 samples	1000	**172**

The same lead both times, and past the boundary the remainder realigned at a shift of one bit with under 1% errors — nothing corrupted, only displaced.

1. The fix

Commit	Change	Why it matters
`3dbd888`	transmit_thread: hand UHD the whole burst in one send() call instead of manually chunking it at get_max_num_samps()	get_max_num_samps() caps a transport packet, not send(). The streamer fragments a larger buffer internally, back to back; fragmenting by hand let the DAC run dry at each hand-off. Two samples of gap is one whole symbol at 2 samples/symbol, so every burst lost a symbol at the first boundary. The preamble sits before that boundary, so sync locked perfectly, the frame decoded byte-perfect up to the gap, and everything after it was displaced by one bit and failed CRC. It presented as a weak link, which is why more gain never helped.

2. What made it findable

Commit	Change	Why it matters
a228e57	Report the first mismatching coded bit, and re-score the tail at shifts -4 to +4	A single BER cannot tell noise from a slip: 27% spread evenly is a dead link, while 0% then 50% averages to the same number and means the opposite. This produced ”bit 911, shift 1” and turned guesswork into arithmetic.
5c51f1b	--tx-spb: set the send-loop chunk size directly	Turned inference into proof. At 1996 samples per call the coded bits first disagreed at 911; at 1000 they first disagreed at 414. Both placed the fault 173 samples into the data field, which is the first send() boundary.
e3e5b6f	New dbpsk_fec_chain_test; per-burst transmit sample accounting	DBPSK with FEC, the only configuration the radio actually uses, had no test coverage at all: the existing front-end test runs coherent schemes without FEC. All 8 combinations pass with zero coded-bit errors, which exonerated the DSP chain and forced the search into the USRP data path.
81dbcc5	Cast idx/tot to int; print received vs computed CRC, and the full payload	idx and tot are uint8_t and streamed as characters, so a correct header (0, 5) printed blank, indistinguishable from a corrupt one. The payload preview also showed only 48 of its 125 bytes.

3. Real bugs fixed along the way

Commit	Change	Why it matters
`90084cd`	Automatic DAC clip guard, plus --tx-scale; clip warning threshold moved from 5.0 to 1.0	The single-carrier path had no amplitude control of its own, and the check only warned above 5.0, so a burst peaking at 1.25 was clipped by the DAC in silence.
`90084cd`	Refuse a coerced sample rate; read back actual vs requested rate, frequency and gain	The code logged what we asked for, never what the hardware returned. A coerced rate drifts a long payload while leaving the short preamble clean.
`7cd8c42`	Drain recv_async_msg: report transmit underflow, sequence and timing errors	The async channel was never read, so a starved DAC transmitted through in silence and looked identical to a healthy transmit.
`89ce8f9`	Report every receive overflow on stdout, with a total at shutdown	It printed once, on stderr, with no total, making it easy to lose the single event that explains an entire run.
`81dbcc5`	Matched filter: treat every detected burst as an independent head	Two consecutive captures of the same length were spliced with stale overlap state. Reproduced and fixed: 64 corrupted samples in range [0..63] before, none after, with a regression test added.
`81dbcc5`	phase_offset_thread: read .size() before std::move, not after	It printed symbols_out=0 on every run, healthy ones included, and was read as evidence of a failing stage.
`d9cddf1`	linktest scripts: proper command-line flags; fixed a pipefail race in the --ber-expected probe	grep -q exits on the first match and SIGPIPEs the binary, so under pipefail the capability test failed at random and BER scoring switched itself off.